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Unlocking Unconventional Targets Using Advanced Fault Connectivity Algorithm

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Abstract

Reservoir compartmentalization poses challenges in unconventional targets between closely spaced wells with different fluid content. Advanced methods are necessary in these settings where traditional seismic interpretations fall behind, relying only on structural analysis. This study presents a refined advanced edge-detecting algorithm for noise reduction, while preserving structural and stratigraphic discontinuities. Using seismic attributes as indicators for hydraulic conductivity, we simulate steady-state flow between horizons utilizing enhanced edge-detection techniques by adjusting iteration weights to improve image clarity, revealing fault connections that were previously undetectable.

This workflow applied to a reprocessed depth migrated seismic data converted to time, revealed lateral separation between zones supporting the hypothesis of hydraulic connectivity across faulted zones. The results offer improved tools for complex subsurface geological structures and provide critical insights for reservoir development and management in unconventional reservoirs.

Keywords: Fault, Connectivity, Seismic, Reservoir

Introduction

Fault detection is a foundational step in understanding subsurface structures and is critical for interpreting fluid migration and accumulation in hydrocarbon reservoirs (both conventional and unconventional), as well as in groundwater aquifers and geothermal systems. In unconventional targets, closely spaced wells often exhibit significant fluid content variation despite their proximity, presenting challenges for reservoir characterization and requiring analysis beyond traditional structural interpretation.

Accurately mapping potential fluid pathways demands a detailed understanding of subsurface faults, which directly influence fluid flow.

Seismic attributes have broad applications in highlighting both structural and stratigraphic features and are instrumental in identifying depositional environments (Chopra and Marfurt, 2007). These attributes are not limited to fault interpretation—they also aid in delineating stratigraphic patterns (Wood, 2000). In particular, edge-detection attributes such as coherence and variance have been widely used to interpret

faults, fractures, and channels. These attributes act as indicators of subsurface compartmentalization, help build robust geological models, and guide fluid flow simulations by offering in-depth structural insights.

A notable contribution in this space was made by [AlBinHassan et al. \(2006\)](#), who developed an advanced edge-preserving algorithm for fault detection in 3D seismic volumes. Their method reduces noise in seismic cubes while preserving structural discontinuities by dividing the seismic volume into blocks and applying directional smoothing that avoids averaging across faults. This results in enhanced coherence attributes that more accurately highlight fault planes, forming a solid basis for deterministic structural interpretation.

However, while this method effectively improves fault imaging, it does not by itself quantify how faults are hydraulically connected. This limitation becomes particularly important in unconventional reservoir settings, where subtle trapping mechanisms and discontinuities play a critical role in controlling fluid flow. These subtle features are often more difficult to detect using conventional multi-trace attributes, yet they can explain dramatic variations in reservoir quality and production observed across short lateral distances. Understanding these features is essential for resolving discrepancies in well performance.

To bridge this gap, Pires de Lima (2017) introduced a method that uses seismic coherence as a proxy for hydraulic conductivity. His approach models faults as flow conduits between bounding horizons and employs a finite-difference scheme to simulate steady-state hydraulic connectivity in three dimensions. This was a major step forward in extending seismic interpretation toward flow-based reservoir modeling. Despite these advances, interpreting subtle fault connections still requires considerable manual effort. Full-scale reservoir simulations can reveal these features but are computationally intensive. To address this, the approach proposed in this study integrates AlBinHassan et al.'s edge-preserving algorithm with deterministic fault connectivity modeling, creating an efficient and scalable solution.

This improved workflow was applied to newly reprocessed depth-migrated seismic data, converted to the time domain and interpreted in the post-stack time migration (PSTM) domain. The results reveal lateral separations and support hypotheses of hydraulic connectivity across terminations. These findings hold meaningful implications for reservoir development, offering a faster alternative to full simulations and enhancing history matching and dynamic data interpretation.

Geological settings

The Late Ordovician glacial event, which occurred approximately 445 million years ago during the Hirnantian stage, represents one of the most significant global climatic transitions in Earth's history. Triggered by a major drop in atmospheric CO₂ levels, this event led to widespread glaciation, predominantly across the supercontinent Gondwana, particularly over what is now North Africa, Arabian plate and parts of South America. The associated sea-level fall had profound implications for sedimentary environments and reservoir deposition patterns worldwide. In shallow marine settings, the glacio-eustatic regression exposed vast continental shelves, leading to subaerial erosion, the development of incised valleys, and the creation of unconformities that now act as key stratigraphic markers. As glaciers waxed and waned, repeated transgressive-regressive cycles reworked sediments, resulting in the deposition of well-sorted, porous sandstones in the Arabian plate.

Furthermore, the abrupt climatic shift contributed to organic-rich Silurian shale deposition in deep basinal settings during transgressive phases, often forming effective source rocks and seals. The interplay between glacial dynamics, sea-level fluctuations, and sediment supply created complex stratigraphic architectures with strong lateral heterogeneity, influencing reservoir quality and connectivity. Understanding the sedimentological and stratigraphic signatures of the Ordovician glacial event is therefore critical in predicting reservoir distribution and performance for hydrocarbon exploration and development.

In the study area, the reservoir forms a thick pile of massive, amalgamated sandstone beds that are interbedded with very thin discontinuous layers of silty sandstone lithologies. Deposition of the sandstones occurred mainly by glacio-marine turbidite fans, which may influence lateral rock property variability in

different turbidite fans. This type of reservoir quality variation may also influence lateral stratigraphic sealing, potentially lateral stratigraphic trapping.

The reservoir in the study area is characterized with low porosity and permeability. This type of reservoir quality requires a rock frack stimulation in order to create an artificial permeability that will allow the release of hydrocarbon for commercial production rates.

The reservoir is a gas-to-condensate bearing zone with a high lateral hydrocarbon variability. This hydrocarbon variability is mostly attributed to the lateral reservoir quality which may accommodate condensates-rich zones in some areas, and gas-rich zones in other areas.

Structural elements are also crucial in forming the hydrocarbon trapping. The study area has undergone a series of tectonics with the Hercynian event, the Hercynian event believed to be the major one forming most of the structural traps. Both structural and stratigraphic traps are believed to have formed prior to the beginning of the charge which is interpreted to have started in the early Triassic era.

Understanding the two components of structural and stratigraphic trapping is crucial for defining the hydrocarbon-rich areas. Focusing only on one element may result in a failure of discovery as it has been experienced. Also, the recent hydrocarbon production data has revealed fluid variations in the same structures, which is attributed to the lateral reservoir variability.

Seismic data can serve as a high-quality proxy for resolving reservoir complexity from the structural understanding as well the lateral reservoir variability. Ultimately, seismic data can be the sweet spotting tool to define the hydrocarbon rich areas within the reservoir in the area of the study.

Methods

The primary aim of this study is to analyze fault connectivity in subsurface regions by leveraging seismic attributes as proxies for hydraulic conductivity. The method assumes that seismic coherence or impedance attributes are proportional to the hydraulic conductivity.

The workflow is comprised of two main steps:

Advanced Edge Detection Using High-Resolution Techniques

Edge detection attributes were generated using the edge-detection algorithm developed by [AlBinHassan et al. \(2006\)](#). This algorithm reduces noise in seismic cubes through use of diffusion operator while preserving structural and subtle discontinuities or edges. The method divides the vicinity of every location in a 3D cube into multiple blocks as the analysis point moves throughout the volume. At interior discontinuity locations with any 3D orientation, the process avoids averaging values across the edge—in order to avoid feature blurring—and instead uses the smoothest neighboring value. The blocks fill all the directions around any point in the 3D cube and hence remove the need for a dip-steering step that usually is required when smoothing seismic amplitude data. The advantage of this 3D smoothing algorithm is its ability to preserve both lateral and vertical edges, producing sharper edge features and allowing the accumulation of stronger discontinuities that indicate the existence of faults.

Fault Connectivity Computation

Building upon the foundations laid by previous studies, our work advances fault connectivity modeling by integrating enhanced edge-preserving seismic attributes with a deterministic hydraulic simulation framework. While prior approaches emphasized optimizing spatial filters or varying window sizes to improve fault imaging, our methodology introduces a distinct strategy that focuses on the iterative behavior of the solver and the impact of iteration weights on image resolution and structural clarity.

The fault connectivity attribute follows the formulation by [Pires de Lima and Marfurt \(2017\)](#), where the flow estimations are computed by using the enhanced edge-detection attributes as proxies for hydraulic conductivity. By simulating fluid flow between horizons, the algorithm highlights faults that connect these horizons while suppressing the response of disconnected faults. The output includes potential hydraulic

head and flow estimates, offering detailed insights into fault connectivity and potential fluid pathways. The formulation of their methodology for simulating hydraulic flow builds on the steady-state three-dimensional saturated flow equation:

$$\frac{\partial}{\partial x}\left(K_x \frac{\partial h}{\partial x}\right) + \frac{\partial}{\partial y}\left(K_y \frac{\partial h}{\partial y}\right) + \frac{\partial}{\partial z}\left(K_z \frac{\partial h}{\partial z}\right) = 0 \quad (1)$$

where K_x , K_y and K_z represent the hydraulic conductivities in the x, y, and z directions, respectively, and h is the hydraulic-head a combination of potential and pressure energy. This equation models the distribution of hydraulic head throughout the medium, assuming that variations in conductivity are linked to seismic reflector misalignments.

An approximate of a one-dimensional fluid flow equation between voxels is solved to estimate the flow rate $Q_{\{i+0.5,j,k\}}$:

$$Q_{\{i+0.5,j,k\}} = K_{x\{i+0.5,j,k\}} \Delta z \Delta y \frac{h_{i+1,j,k} - h_{i,j,k}}{\Delta x} \quad (2)$$

where $Q_{\{i+0.5,j,k\}}$ is the volumetric flow rate through the face between voxels $\{i, j, k\}$ and $\{i + 1, j, k\}$, and $K_{x\{i+0.5,j,k\}}$ is the hydraulic conductivity along the row between these nodes. The approximation is extended to three-dimensional (3D) space for our problem, where flow occurs in all three directions. This equation forms part of a linear system that is iteratively solved using the weighted Jacobi method. A key refinement introduced in this study lies in how the hydraulic simulation framework was implemented and tuned. While the original algorithm proposed by Pires de Lima and Marfurt (2017) emphasized solving the steady-state flow equation to estimate head and flow fields between horizons, our implementation revisited how the iterative solver behaves when applied to high-resolution edge-detection-derived conductivity volumes. Unlike impedance-based approaches that often require thousands of iterations to achieve stability, we found that limiting the solver to 500 iterations yielded stable, interpretable results across various datasets. The clarity of flow pathways and fault connectivity patterns plateaued beyond this point, with minimal improvement observed at higher iteration counts.

Our implementation also highlighted that the weighting factor (ω) used in the iterative scheme had significantly less influence on the resulting quality than initially expected. We experimented with a range of weights and found that, for conductivity fields derived from enhanced edge-detection attributes, the algorithm was robust to weight variation, reinforcing the idea that attribute sharpness and data conditioning play a more important role than fine-tuning numerical solver parameters. To further improve efficiency, we exported the hydraulic solver to GPUs using CuPy, enabling rapid processing of large 3D seismic datasets. With this parallelized design, the edge-detection phase was completed on a 40-CPU cluster in under 4 hours, while the hydraulic connectivity computation itself was executed in less than 1 hour on a single GPU. This level of performance makes the method well-suited for integration into operational workflows, where turnaround time and scalability are critical.

Together, these enhancements demonstrate that the algorithm can be made both computationally efficient and geologically meaningful without relying on extensive iteration counts or solver parameter tuning. By leveraging high-quality edge-based attributes and optimizing the solver architecture, we achieved detailed insights into fault connectivity at scale, offering a practical and interpretable solution for subsurface analysis.

Results and Discussion

The method was applied to an unconventional reservoir where two closely-spaced wells showed significant differences in fluid content, presenting challenges for accurate reservoir characterization and development planning. Traditional seismic interpretation failed to identify the discontinuities responsible for this variation, leading to uncertainty in understanding subsurface fluid distribution. By applying the advanced edge-detection algorithm, we revealed clearer fault patterns in the area of interest, enhancing the

visibility of structural features that likely influence fluid compartmentalization. The algorithm produced sharper edge definitions and emphasized areas of strong discontinuity, highlighting potential fault zones with greater clarity. This improved imaging provided key insights into the subsurface, allowing us to identify fault segments responsible for lateral separation between the two zones. Additionally, steady-state flow simulation and flux analysis using fault connectivity attributes highlighted the hydraulic linkage between faulted zones above and below the reservoir. These results support the interpretation that lateral fluid variation is structurally controlled, explaining the observed differences between the wells. This understanding is critical for optimizing well placement, refining production strategies, and improving resource estimation. By accurately modeling fault connectivity, we enhance our ability to predict fluid migration and recovery potential within the reservoir.

Adjusting the iteration weight during edge detection proved essential for enhancing fault visualization. By consistently applying 1000 iterations, we achieved an effective balance between suppressing noise and preserving meaningful geological edges. This higher iteration significantly improved the definition of fault boundaries, allowing the identification of subtle discontinuities that may serve as fluid conduits or barriers. The sensitivity of the results to iteration weight highlights its importance as a key parameter in edge-detection workflows. Careful calibration of this factor, tailored to the specific dataset and geological context, enhances the method's adaptability across different subsurface environments. Importantly, optimizing the iteration weight helps avoid over-smoothing, which can obscure critical features, while still effectively reducing random noise that could lead to misinterpretation. This balance is crucial for maintaining geological integrity and ensuring accurate fault mapping in complex settings.

The resulting fault connectivity cube enabled detailed structural mapping, as shown in [Figure 1](#). The highlighted fault patterns allowed for precise segmentation of the area, with individual segments aligning with variations in pressure and fluid data. Some discontinuities were interpreted as structural features consistent with core observations. In unconventional reservoirs with low permeability, even minor structural discontinuities can act as barriers, compartmentalizing fluid distribution and reservoir properties. Integrating geological data with fault connectivity analysis supports the interpretation that these structural breaks are responsible for the observed differences in fluid content. The layer offset between the wells appears to function as a barrier, effectively dividing the reservoir into two distinct fluid compartments ([Figure 2](#)). [Figure 3](#) shows the same cross section in the fault connectivity attributes highlighting the same subtle feature. This understanding is critical for guiding reservoir development and optimizing production strategies in complex subsurface environments.

Compared to conventional edge-detection techniques, our method delivers superior fault resolution and connectivity analysis. By integrating an advanced edge-preserving smoothing algorithm with fault connectivity computations, this approach marks a meaningful step forward in seismic interpretation. While Pires de Lima (2017) demonstrated the effectiveness of using seismic coherence as a proxy for hydraulic conductivity across various datasets, our study builds on this foundation by introducing iteration weight adjustments to further refine fault imaging. This enhancement enables the detection of subtle geological features that may be missed by traditional methods, resulting in a more detailed and accurate representation of subsurface structures. Ultimately, this approach bridges the gap between seismic interpretation and hydraulic modeling, offering a more cohesive and robust framework for subsurface analysis.

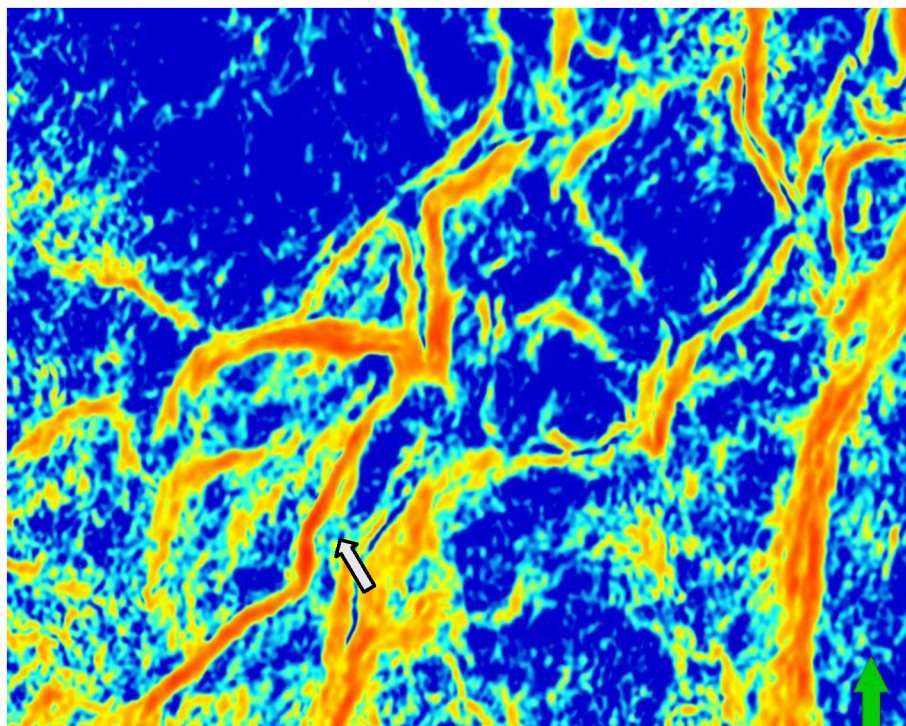


Figure 1—Time slice from the fault connectivity cube. High amplitude signatures are indicative of fault structures, not all were captured in conventional attributes. Low relief signatures indicate subtle features the could represent small faults structures. The arrow highlights the location of the subtle feature between the two wells.

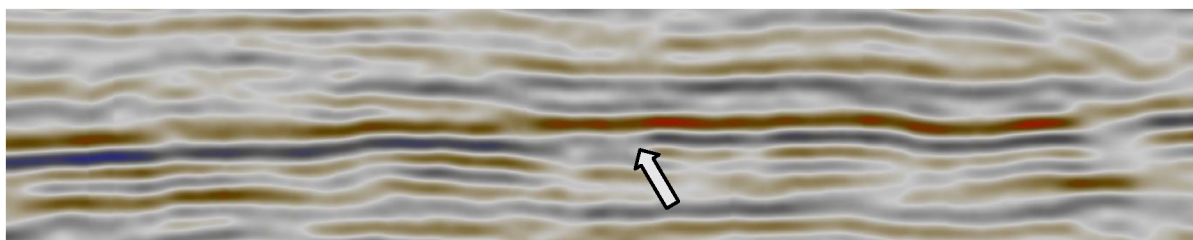


Figure 2—Seismic amplitude section crossing the two wells under investigation. The arrow shows the layer with the subtle feature.

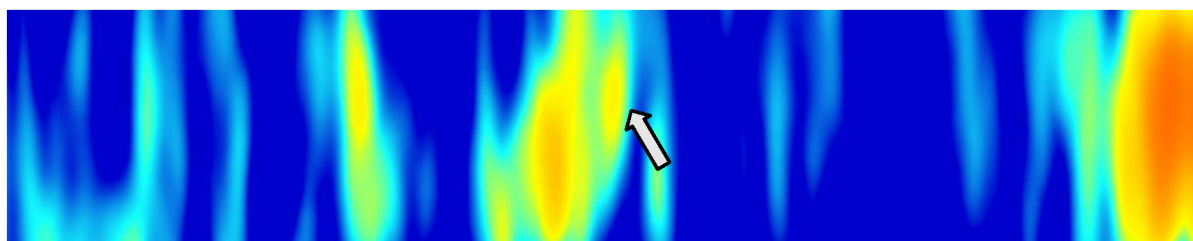


Figure 3—Seismic section from the fault connectivity cube. The arrow shows the location of the subtle feature which is captured on the fault connectivity as a low relief signature.

Although the method shows clear advancements in fault detection and connectivity analysis, several limitations should be considered. Its effectiveness depends heavily on the quality and resolution of seismic data; areas with poor data remain challenging, as noise and artifacts can compromise edge detection and, by extension, fault connectivity results. Additionally, the method relies on the assumption that seismic attributes correlate directly with hydraulic conductivity—a relationship that may not consistently apply across varying geological settings. Factors such as lithology, fluid content, and pressure conditions can

influence this correlation. Therefore, further validation using well data (e.g., core samples and logs) and complementary geophysical measurements is essential for refining interpretations and calibrating model parameters. The method may also be less effective in geologically-complex areas with intersecting fault systems or highly-intricate fault geometries, where additional structural and stratigraphic data are needed for accurate subsurface characterization. Despite these limitations, the approach remains a valuable tool for improving fault imaging and fluid flow interpretation, supporting better-informed decisions in exploration and reservoir development.

Conclusion

This study introduces an improved method for analyzing fault connectivity in subsurface environments, with a focus on unconventional reservoirs exhibiting lateral fluid variability. By refining edge-detection techniques and tuning iteration weights, we advanced existing methodologies to uncover previously undetected connections between faulted zones. This enhanced imaging provided clearer insights into potential fluid pathways and lateral compartmentalization, which are often missed by traditional seismic interpretation. The use of fault connectivity attributes further demonstrated the ability to detect subtle zone separations, offering a more accurate understanding of reservoir structure. The approach was computationally efficient, with parallel processing enabling rapid analysis, making it practical for industry settings where timely subsurface evaluation is critical. These advancements have significant implications for fluid flow modeling, reservoir characterization, and resource management. By serving as a rapid alternative to conventional reservoir simulations, the method supports better prediction of fluid migration, identification of flow barriers, and more informed exploration and development decisions. This integration of geophysical analysis and computational optimization presents a valuable tool for geoscientists and reservoir engineers.

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